Lecture 13 - Oct 12

Recall Lecture 11 - Oct 5 > Definition 2 (Binary relation)

Review and corrections

A corrected definition of total order:

DEFINITION (LINEAR/TOTAL ORDER).

A **linear order/total order**I is a partial order where each pair of elements in X is comparable.

DEFINITION (EQUIVALENT RELATION).

a relation is an **equivalent relation** if its reflexive, symmetric, and transitive. Such a relation is denoted with $\sim$

REMARK (EQUIVALENCE RELATION).

If $\sim$ is an equivalent relation on a set X, then $\{[x] : x < X\}$ is a partition of X. Similarly, any partition of X yields an equivalent relation on X.

An equivalent class is defined as the following:

$$\begin{gathered}\forall x \in X \\ [x] := \{y \in X | x \sim y\} \\ [x] \neq \emptyset \text{ since } x \in [x] \\ [x] \cap [y] = \emptyset \text{ if } x \nsim y\end{gathered}$$

All the following are defined for a poset (partial order set) $P = (X, \leq)$:

- chain, antichain, maximal chain, maximal antichain

X can be partitioned into chains or antichains, meaning, all elements of X are in exactly one chain/antichain.

Let $P = (X, \leq)$ be a poset, where X is finite. The largest antichain in P has size m and the largest chain in P has size n. Let $C_1, \ldots, C_r$ be a chain partition of X and $A_1, \ldots, A_s$ be an antichain partition of X.

Observation:

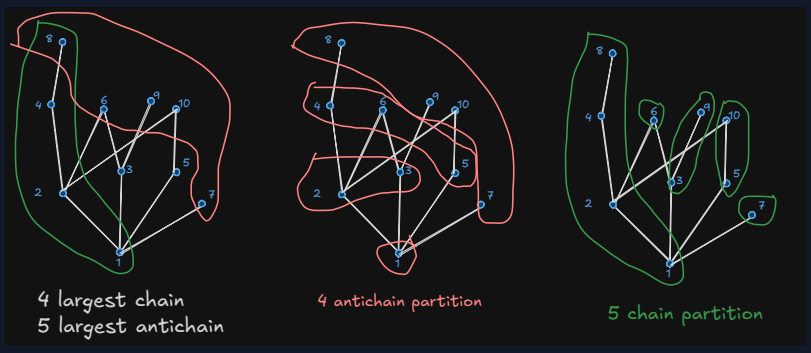
1. $r \geq m$, since $|C_i \cap A| \leq 1$
2. $s \geq n$, since $|C \cap A_i| \leq 1$

Practice Example

Given poset $P = ([10], |)$

1. Draw the Hasse Diagram
2. Find the largest chain and antichain

3. partition X into as few chain as possible antichains



THEOREM (DILWORTH THEOREM).

Let $P = (X, \leq)$ be a finite poset such that the largest antichain has size m and the largest chain has size n. Then, X can be partitined into exactly m chains and n antichains.

True of False:

1. Let $(X, \leq)$ be a poset and let A be the set of all minimal elements of X. Then for every $y \in X - A$, there exists $x \in A$ such that $x \leq y$. -> **True**.

Proof.@Theorem 4 (Dilworth Theorem)

We need to construct an antichain partition of X with exactly n antichains.

Let A_1 be the set of all minimal elements of $P = (X, \leq)$.

A_2 be the set of all minimal elements of the poset $(X_1, \leq), X_1 = X - A_1$

...

Remark: This algorithm eventually stops since it's a **finite** poset.

We continue until the last p such that $X_p = X_{p-1} - A_{p-1} \neq \emptyset, X_{p+1} = \emptyset$. Let $X_p = A_p$.

It remains to show $p = n$. It is known that $p \geq n$.

By the true of false question, $\exists x_i \in A_i$ such that $x_1 \leq x_2 \leq \cdots \leq x_p$, implying $x_1, \ldots, x_p$ is a chain in P, implying $p \leq n$.

□

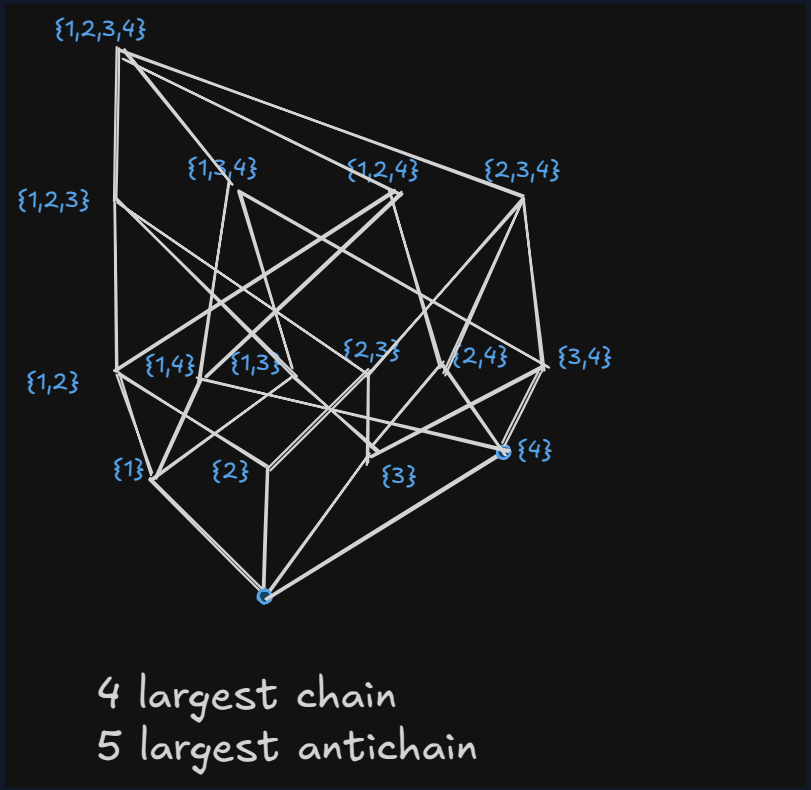
Homework problem: proof of the true or false question -> could be on the test!!!

Example: Type of problem that could appear on the exam

$P = (\mathcal{P}([4]), \subseteq)$

1. Draw the Hassel diagram
2. Find all maximal chains and the largest chain
3. Partition $\mathcal{P}([4])$ into as few antichains as possible
4. Find all maximal antichains

Diagram:


{1,2,3,4}
{1,3,4}
{1,2,4}
{2,3,4}
{1,2,3}
{1,4}
{1,3}
{2,3}
{2,4}
{3,4}
{1,2}
{1}
{2}
{3}
{4}
4 largest chain
5 largest antichain